

ual:

# Coding Three

Creative Robotics

Additional Brief on Element 2: Mini Project

# Learning Outcomes

- **LO1** Understand how machine learning works in practice (**Knowledge**)
- **LO2** Demonstrate practical skills in computer programming for use in Computer Vision contexts (**Process**)
- **LO3** Use existing frameworks and examples to create computational systems that make use of Computer Vision approaches relevant to robotics (**Realisation**)

# Mini Project

- You will work on a Mini Project of your own throughout the Unit
- A Mini Project should be:
  - A small creative project made with computational systems and primarily based on Python code
  - An application of *at least* one computer vision model that is based on machine learning in creative practice
  - Made relevant to robotics (this does **not** necessarily need to include a component for robotics or physical computing in the project)

# Mini Project

- There will be sessions during the Unit that support the progress of your Mini Project, the activities of the sessions include but are not limited to:
  - In-class planning and designing
  - In-class making
  - Feedback from peers and tutor(s)
  - Surgery: Tweaking and improving code or the project's operation with a tutor

# Mini Project Submission

- For Assessment, at the end of the Unit, you will submit:
  - Digitally presented outcomes of your Mini Project, its documentation of the process of development, in a digitally edited document titled *Element 2 Submission*.
  - The *Element 2 Submission* document should contain images, texts and optionally links, videos, audio and any other relevant media files to explain your project
  - Within your *Element 2 Submission* document, it should also include your own GitHub Repository link that contains **all the coding files related to this project**

# Mini Project Submission

- In the *Element 2 Submission* document, please provide
  - **Documentation of the progress and outcomes** of the Mini project
  - **A link containing your GitHub Repository with all the coding files related to this project**
    - You don't need to include the ML model you have used for your project, *however*, you should reference to them in your write up so that we can see what ML model you chose for your work and your rationale on choosing this particular model
  - **Text in two parts:**
    - Project description (400 words maximum)
    - Reflective evaluation of the project development (500 words maximum)

# Mini Project Submission

- Documentation of the outcomes of the Mini project should provide:
  - A clear explanation of how your Mini Project operates and functions
  - The progression of your Mini Project from conception to finish
  - You can use images and videos with annotations explaining what they do/represent
  - It should also present the trajectory of how you developed your project from initial ideas or prototypes to their variations to the final version
  - Keep your annotations brief: 50-80 words maximum per annotation, unless the use of more words is reasonably appropriate. For instance, to explain a complex mechanism.

# Project Description

## ■ Should describe:

- What you made
- Which technical approaches are used
- How machine learning is used in the project
- Why you chose to employ particular methods (if relevant)
- How the project can be seen as relevant **to the context of robotics** i.e. either clarified through your explanation of how the chosen computer vision model can operate with a particular robotic system and its functionalities, or by demonstratively integrating robotic components with the computer vision used (the latter is an optional approach)

# Reflective Evaluation

- Should describe **how you researched and experimented with different tools and methods** related to computer vision and why you chose the final methods from different options
- To work on this, you might start with questions such as:
  - What went well and what did not go well through your experimentation or tests
  - What have you managed to improve and how?
  - How effectively did the chosen machine learning functions work in your project?
  - How relevant are your project and computer vision approaches to the context of (creative) robotics?



**Questions?**

